A MACHINE-CHECKED ATLAS OF RIEMANN-HYPOTHESIS EQUIVALENCES AND NO-GO CERTIFICATES

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Abstract. We describe `jensen-ladder`, a Lean 4 library that formalizes, against `mathlib`, the *logical geometry* of the Riemann Hypothesis (RH): the network of equivalent reformulations and the structural obstructions that any proof must respect. The library contains 128 source modules; the full `lakebuild` is green (8602 compiled jobs); the tree contains no `sorry` and no added `axiom`, and the headline results depend only on the three standard `mathlib` axioms `[propext,Classical.choice,Quot.sound]`. Three things are mechanized. First, a short, unconditional *reduction spine* that proves `mathlib`'s official `RiemannHypothesis` equivalent to the reality of the *regular* zeros of the completed zeta function in a symmetric variable, with the Gamma-factor exceptional set and the pole handled exactly, and the functional equation realized as a reflection symmetry. Second, an *equivalence lattice*: RH is proved equivalent — in the kernel, axiom-clean — to the existence of any one of a family of abstract *faithful carriers* (Hodge-index, arithmetic-site, geometric-square-root, Deninger-dictionary, moment-problem, spectral-realization, Morse-index-zero, Fredholm-squared, . . .), each an interface that records *what kind of object a proof must supply*, and each shipped with its own *falsifier* (a single off-line zero refutes it) and with *non-supply* lemmas certifying that naive (finite, diagnostic, fake) candidates do not instantiate it. Third, a suite of finite, fully proved *no-go certificates* that rule out entire classes of naive strategy — the Davenport–Heilbronn fake-family obstruction, the finite-carrier obstruction, the failure of one-sided prime-local positivity, the multiplicativity-visibility obstructions, a scattering-parity obstruction, and the no-margin / resolution-wall obstruction. The novel contribution is not a new theorem about ζ: it is the machine-checking itself. The RH "wall" — known informally as a scattering of equivalences and folklore obstructions — becomes one axiom-clean, mutually consistent, kernel-enforced formal artifact. **The library does not prove RH, and does not claim to:** every load-bearing carrier hypothesis is left unproven, several are proved *equivalent* to RH, and the genuinely open input is named explicitly.

1. Introduction

What is formalized, and what is not. The Riemann Hypothesis asserts that the nontrivial zeros of the Riemann zeta function lie on the line $\operatorname{Re} s = \frac{1}{2}$. In `mathlib` it is the proposition

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`RiemannHypothesis:∀s,riemannZetas=0→(¬∃n,s=-2*(n+1))→s≠1→s.re=1/2.`

This paper is *not* a proof of RH. It is a description of a Lean 4 library that formalizes the *structure* of the problem: the equivalent forms RH can take, and the obstructions that constrain any proof. We call this structure the *logical geometry of the wall.* Informally this geometry is well known — RH has dozens of published equivalent reformulations [6, 7, 8], and the function-field analogue (Weil, Deligne) makes the shape of a hypothetical proof a recurring subject of speculation [9, 10]. What has *not* existed is a single machine-checked artifact in which these reformulations are stated precisely against a common formal kernel, proved equivalent to one another, and confronted with the obstructions in the same logical universe. That artifact is the contribution here.

Three layers. The library, `jensen-ladder`, has three layers.

(1) **The reduction spine** (§2). A handful of unconditional, axiom-clean theorems carry `mathlib`'s `RiemannHypothesis` to a clean working endpoint: the reality of the *regular* zeros of the completed zeta function Λ written in the symmetric variable $\Xi(z) = \Lambda(\frac{1}{2} + iz)$. The functional equation becomes evenness of Ξ; the trivial-zero / Gamma-factor exceptional set and the pole at $s = 1$ are filtered exactly.

(2) **The equivalence lattice** (§3). RH is proved equivalent, in the kernel, to the existence of any one of a family of abstract *faithful carriers*. Each carrier is a Lean `structure` whose "faithful rows" would suffice to prove RH (`riemannHypothesis_of_faithfulRows`); each ships with a falsifier (`not_faithfulRows_of_nonrealRegularXiZero`); the carriers are proved mutually equivalent, forming a lattice whose top is RH.

(3) **The no-go certificates** (§4). Finite, fully proved theorems that exclude naive proof strategies: that a single Euler-product coefficient law is the only separator (the Davenport–Heilbronn fake gate), that no finite spectrum can host infinitely many zeros, that one-sided prime-local positivity fails, that finite "visibility" cannot force multiplicativity, that a scattering ratio cannot equal Ξ, and that no margin/floor mechanism survives (the resolution wall).

The point of mechanizing this. A skeptic may object that an equivalence "RH $\iff$ (an object exists that would prove RH)" is logically shallow. The objection is partly correct and we take it seriously throughout (§7). The value of the mechanization is fourfold and is *not* mathematical depth of any single equivalence:

- *Faithful bookkeeping.* The reduction targets are stated as precise Lean types, not prose. "What object would suffice" becomes a checkable `structure`; "what is still open" becomes a named, unproven hypothesis of known status.
- *Kernel-enforced consistency.* The carriers are proved *equivalent to one another*, so the routes cannot secretly disagree about what RH needs; the falsifier direction is uniform across all of them; the no-gos compose with the lattice in the same logical universe.
- *The no-gos are genuine theorems.* Unlike the equivalences, the no-go certificates are finite, self-contained, unconditional results — a machine-checked census of what cannot work.
- *Honest separation.* The library distinguishes, at the type level, proved analytic content about ζ from abstract interfaces that merely *record* an open problem. Every module that consumes an unconstructed object says so in its doc-string.

Relation to Theorem M. A separate, self-contained result — "Theorem M", the unconditional hyperbolicity of the critical Laguerre family Ψ_d [5] — is a proved *input* to one route of the lattice (the model-to-Ξ transfer). We emphasize, as that work does and as every relevant module here repeats: **Theorem M is proven, but Theorem M does not prove RH by itself.** The present library makes precise exactly which additional, unproven input would be required.

Verification at a glance. The toolchain is pinned (`leanprover/lean4:v 4.30.0`, `mathlib v4.30.0`). The complete check is

`cdformal&&lakebuild` (green, 128 modules, 8602 jobs)

followed by `scripts/check_axioms.sh`, which prints the axiom dependency of a curated set of 16 headline theorems spanning all three layers and confirms each is exactly `[propext,Classical.choice,Quot.sound]` with zero `sorry`s (§8).

2. The reduction spine

The endpoint of the analytic program is most naturally stated about the *completed* zeta function $\Lambda =$ `completedRiemannZeta` and its symmetric variable. Module `RHReduction` formalizes the bridge from `mathlib`'s `RiemannHypothesis` to that endpoint, unconditionally.

The Gamma-factor bookkeeping. `mathlib` provides $\zeta(s) = \Lambda(s)/\Gamma_{\mathbb{R}}(s)$ for $s \neq 0$, where $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ vanishes exactly on $\{0, -2, -4, \dots\}$. These are precisely the points excluded from the RH quantifier (the trivial zeros, together with $s = 0$, where $\zeta(0) = -\frac{1}{2} \neq 0$). Hence at every point in scope of the RH quantifier $\Gamma_{\mathbb{R}}(s) \neq 0$, so a zero of ζ there is a zero of Λ. Define a *regular* zero to be one with $\Gamma_{\mathbb{R}}(s) \neq 0$ and $s \neq 1$. The module proves both directions and packages them as an equivalence.

Theorem 2.1 (`riemannHypothesis_iff_completedZeta_regular_zeros_on_line`). *`RiemannHypothesis` holds if and only if every regular zero s of Λ has* $\operatorname{Re} s = \frac{1}{2}$.

The symmetric variable and the functional equation. Set $\Xi(z) = \Lambda(\frac{1}{2} + iz)$ (`riemannXi`). The change of variables $s = \frac{1}{2} + iz$ sends $\operatorname{Re} s = \frac{1}{2}$ to $\Im z = 0$, and a regular zero of Λ to a *regular Ξ-zero* (`riemannXiRegularZero`: the Ξ-zero condition together with the two regularity filters). The functional equation $\Lambda(1 - s) = \Lambda(s)$ becomes the evenness of Ξ.

Theorem 2.2 (`riemannXi_even`, `riemannXiRegularZero_neg`). $\Xi(-z) = \Xi(z)$ *for all $z \in \mathbb{C}$, and the regular-zero predicate is invariant under $z \mapsto -z$. The fixed-point set of this reflection is the real axis $\Im z = 0$, i.e. the critical line.*

The second statement uses one nontrivial `mathlib` input (nonvanishing of ζ on $\operatorname{Re} s \geq 1$) to keep the regularity filters symmetric; it is the formal heart of "functional equation = reflection about the critical line". Composing the change of variables with the Gamma-factor bridge yields the spine's headline equivalence.

Theorem 2.3 (`riemannHypothesis_iff_regular_riemannXi_zeros_real`). *`RiemannHypothesis` $\iff$ every regular Ξ-zero is real ($\Im z = 0$).*

All of §2 is unconditional and axiom-clean. It is the one place in the library where "RH" is connected to a tractable analytic object; every later layer targets the right-hand side of Theorem 2.3.

3. The equivalence lattice

The carrier pattern. Each route is organized around an abstract `structure` — a *carrier* — whose fields are the rows a proof along that route would have to supply. The carrier is deliberately abstract: it does *not* construct the geometric or spectral object, only names the properties that object must have. Every carrier module instantiates the same three-part pattern:

(1) **sufficiency:** `riemannHypothesis_of_faithfulRows` — the faithful rows imply RH (via Theorem 2.3);
(2) **falsifier:** `not_faithfulRows_of_nonrealRegularXiZero` — a single off-line regular Ξ-zero refutes the rows;
(3) **non-supply:** lemmas of the form `X_do_not_supply_Y` exhibiting an explicit *diagnostic* carrier (e.g. empty spectrum, fake coefficient stream) that satisfies the cheap rows but provably not the faithful ones.

The non-supply lemmas are what keep the pattern honest: they certify that the carrier's faithful row is not vacuously satisfiable, so the equivalence does not collapse to a triviality at the cheap end.

The lattice. The carriers are then linked by kernel-checked `iff`s into a single lattice whose top element is RH. Representative equivalences (all axiom-clean):

$$\begin{aligned}\mathrm{RH} &\iff \exists \text{ a faithful Hodge-index carrier}\\ &\iff \exists \text{ a faithful arithmetic-site carrier}\\ &\iff \exists \text{ a faithful geometric-square-root carrier}\\ &\iff \exists \text{ a faithful Fredholm-squared carrier}\\ &\iff \exists \text{ a regular spectral realization}\\ &\iff \text{no negative Morse modes (Morse index 0)},\end{aligned}$$

with the corresponding module and theorem names collected in Table 1. The intermediate equivalences are proved *directly between carriers* (e.g. `hasFaithfulHodgeIndexCarrier_iff_hasFaithfulArithmeticSiteCarrier`, `hasFaithfulGeometricSquareRootCarrier_iff_hasFaithfulHodgeIndexCarrier`), so the lattice is internally consistent: every route agrees, in the kernel, on what object RH requires. Table 1 lists the carrier modules.

A worked carrier: the finite Hodge-to-Weil bridge. To show the pattern carries genuine finite content (not only abstract plumbing), consider `HodgeWeilBridge`. For a finite semilocal Weil matrix D (archimedean part minus prime/local part) it defines the quadratic form $Q_D(v) = \sum_{i,j} v_i\,(\texttt{entry}\,D\,i\,j)\,v_j$ and proves the exact split

$$Q_D(v) \;=\; Q_D^{\mathrm{arch}}(v) \;-\; \sum_q Q_D^{\mathrm{prime},q}(v)$$

(`quadraticForm_eq_arch_sub_sum_localPrimeQuadraticForm`). It then defines an abstract `FiniteHodgeRealization` — a map from finite vectors to a `DivisorClass` with an intersection form, an abstract `primitive` predicate, the Hodge-index row (primitive self-intersections ≤ 0), and the identification $Q_D(v) = -\langle \mathrm{realize}\ v, \mathrm{realize}\ v\rangle$ — and proves that any such realization makes Q_D positive semidefinite, while a single negative vector rules out every realization. The doc-string states the boundary precisely: "This file does not construct the arithmetic surface, divisor classes, intersection theory, trace identity, semilocal Connes operator, convergence theorem, or RH. It only records the finite algebraic bridge that such a construction must instantiate." That is the honest shape of the whole lattice: real finite algebra, an explicitly abstract geometric interface, and a named open problem.

4. The no-go certificates

The second class of results is mathematically self-contained: finite, unconditional theorems that rule out naive strategy classes. These are the machine-checked "wall". Table 2 lists them; we expand the two most structurally important.

Module	Carrier / equivalence (open content = existence of the faithful rows)
`HodgeIndexCarrier`	RH $\iff$ faithful Hodge-index carrier $\iff$ arithmetic-site carrier
`ArithmeticSiteCarrier`	RH $\iff$ faithful arithmetic-site carrier (Lefschetz/Weil-trace rows)
`GeometricSquareRootCarrier`	nonnegative Weil form $\iff$ abstract square root; $\iff$ Hodge-index carrier
`HodgeWeilBridge`	finite Hodge realization $\Rightarrow$ Weil form PSD; negative vector $\Rightarrow$ no realization
`PrimitiveHodgeWeilEngine/Calibration`	primitive Hodge–Riemann engine for finite Weil forms
`DeningerCarrier`	RH $\iff$ polarized faithful dictionary (flow / determinant interface)
`DeningerFlowUnitary`, `FriedDeningerTorsionCarrier`	flow-unitarity / torsion variants of the Deninger dictionary
`MomentProblemCarrier`	RH $\iff$ faithful moment-problem carrier $\iff$ spectral realization
`SpectralRealization`	RH $\iff$ $\exists$ (regular) spectral realization of the Ξ-zeros
`SpectralReflection`	RH $\iff$ $\exists$ reflected (FE-symmetric) regular spectral realization
`ComplexSpectralDeterminantCarrier`, `UnitaryCokernelCarrier`	complex-determinant / unitary-cokernel realizations
`FredholmSquaredCarrier`	RH $\iff$ faithful Fredholm-squared carrier $\iff$ nonnegative squared support
`SquaredVariablePullback`	reality from nonnegative squared support / approximation
`PrimeCepstrumHankelCarrier`	prime-cepstrum Hankel realization (feeds Fredholm-squared)
`MorseCriterion`	RH $\iff$ no negative modes $\iff$ nonnegative spectral bottom
`MorseDeningerBridge`	no negative modes $\iff$ polarized faithful dictionary $\iff$ RH
`BochnerTopRoute`	RH $\iff$ nonpositive "top" (Bochner simple-top calibration)
`MagicInterpolationBridge`	magic-interpolation data $\Rightarrow$ RH; off-line zero $\Rightarrow$ no magic rows (sufficiency + falsifier; Viazovska-style target)
`ModelToXiTransfer`, `PolyaSchurTransfer`	model endpoint (Theorem M) + transfer row + Jensen gate $\Rightarrow$ RH
`TruncationLimitCarrier`, `SpectralPollutionFreeLimit`	exhaustive / pollution-free limit carriers
`CVSSpectralRoute` (+ `CVS*`, `CCMGroundState*`, `OddMorseCertificate`)	Connes–van Suijlekom finite-scale ground states + convergence $\Rightarrow$ RH
`DeterminantHurwitzRoute`, `SquaredDeterminant*`, `DiagonalFredholmProduct`	`det_reg` $\to \Xi$ squared-determinant route

TABLE 1. The equivalence lattice. In every row the *open content* is the existence of the faithful rows; the equivalence and the falsifier are proved.

Multiplicativity is the only separator (Davenport–Heilbronn). A recurring failure mode is any argument that uses only the functional equation, reality, and density/coefficient information — because the Davenport–Heilbronn function has all of those and yet has zeros off the critical line.

Module `DHMultiplicityFakeGate` formalizes the finite arithmetic core that distinguishes ζ from such fakes: a single normalized Euler-product coefficient stream must be (coprime) multiplicative, and the Davenport–Heilbronn residue pattern already violates this at the coprime pair $(2,3)$.

Theorem 4.1 (`not_coprimeMultiplicative_of_DHPattern236`). *If* $a(1) = 1$, $a(2) = \kappa$, $a(3) = -\kappa$, $a(6) = 1$, *then* a *is not coprime multiplicative — multiplicativity would force* $a(6) = a(2)a(3) = -\kappa^2 \neq 1$.

The companion modules `FiniteMultiplicityVisibilityNoGo`, `FiniteMixedLogVisibilityNoGo`, `FiniteEulerProductMixedLogGate` sharpen this into a *visibility* statement: a finite "visible predicate" (any finite window of coefficient data) cannot force coprime multiplicativity, because a product-spike pattern agrees with the unit stream on the window yet violates the law off it. This is the formal version of "the Euler product is the unique separator, and no finite amount of it suffices".

No margin (the resolution wall). RH sits at an *exact* threshold: empirically the minimal normalized zero-gap tends to 0 (the Lehmer phenomenon), so no proof can rely on a uniform positive margin/floor. Module `ResolutionWall` (with `BAH1Quadratic`) formalizes the contradiction.

Theorem 4.2 (`not_forall_certificate_margin_of_uniform_floor_arbitrarily_small_margins`). *If certificate margins are bounded below by a fixed positive floor, but the required margins are arbitrarily small, no certificate can meet the margin at every index — a contradiction. Hence a margin/floor mechanism cannot certify the critical (marginless) inequality.*

This is the formal counterpart of the analytic "no-floor" fact and explains, inside the kernel, why soft-margin and single-positive-functional methods are excluded. `CVSKatoSmallNoGo` reaches the same conclusion on the spectral side: Kato-smallness fails precisely when the spectral gap collapses.

5. Finite positivity algebra and unconditional analytic bricks

Two further families of modules carry genuine mathematical content below the abstract interfaces.

Finite positivity algebra. A toolkit of finite linear algebra realizes the "square-root/domination" mechanics the positivity routes need: `HodgeSignatureBabyCarrier`, `PrimeDominationBabyCarrier`, `GraphLaplacianBabyCarrier`, `RowDiagonalDominationCarrier`, `GlobalDominationReduction`, `SchurStarDominationCarrier/AssemblyCarrier`, and the finite-rank `CCM*` algebra (`CCMFiniteWeil`, `CCMRankOne`, `CCMQuotient`, `CCMFrobeniusBound`, `CCMPerturbationBound`, `CCMFiniteKatoBridge`). These prove, e.g., that an off-diagonal coupling dominated by the diagonal makes a completed block positive semidefinite, and that a rooted incidence assembly's energy is an exact square. The `Chiral*` modules formalize a finite chiral-Dirac

Module	What is proved (finite, unconditional)
`DHMultiplicityFakeGate`	the Davenport–Heilbronn pattern is not (coprime/totally) multiplicative
`FiniteMultiplicityVisibilityNoGo`	a finite visible predicate cannot force coprime multiplicativity
`FiniteMixedLogVisibilityNoGo`, `FiniteEulerProductMixedLogGate`	finite mixed-log data cannot force the product (Euler-coupling) relations
`FiniteCarrierNoGo`	a finite spectrum cannot represent infinitely many distinct zeros
`PrimeLocalNoGo`	one-sided prime-local kernel is negative below threshold $\Rightarrow$ local positivity fails
`PrimeLocalHodgeNoGo`	one-sided prime-local data admits a negative Hodge vector
`ScatteringParityNoGo`	a scattering ratio (even log-derivative) cannot equal Ξ (odd) — closes the ratio class
`ResolutionWall`, `BAH1Quadratic`	uniform floor + arbitrarily small margins $\Rightarrow$ contradiction (no margin)
`CVSKatoSmallNoGo`	Kato-smallness fails under critical gap collapse
`AdversarialSignTwistNoGo`	an adversarial sign-twist forces a negative direction unless budget $\leq$ archimedean
`AbsoluteBudgetFakeFamilyBlindness`	absolute-budget certificates are blind to sign-retwisted fake families
`DeterminantNormalizationNoGo`	post-hoc scalar normalization cannot repair `det_reg`
`SpectralFaithfulnessGap`	an off-axis regular zero $\Rightarrow$ the squared spectrum is not faithful
`SchurStarSharpness`	archimedean < response budget $\Rightarrow$ an explicit negative form (sharpness)

TABLE 2. The no-go certificates: a machine-checked census of strategies that cannot work.

perfect-square route (PSD Hankel moments, Stieltjes trace identities) with explicit "diagnostic rows do not supply live rows" honesty lemmas.

Unconditional analytic bricks. Several modules prove substantive, RH-independent facts about the completed zeta function, some filling gaps not in `mathlib`:

- `RvMXiEntire` — the entire completion $\texttt{xiE}(s) = \frac{1}{2}(s(s-1)\,\texttt{completedRiemannZeta}_0(s) + 1)$ is entire, with nonnegative divisor, and $\texttt{xiE}(s) = 0 \iff \zeta(s) = 0$ on the critical strip (`xiE_zero_iff_zeta_zero`); the Riemann–von Mangoldt foundation.
- `ZeroCountingN` — the zero-counting function $N(T)$ as a finite, monotone count.
- `XiOrderBound`, `XiZeroCounting`, `W1DensityNevanlinna`, `CompletedZetaStripBound` — order-1 growth bounds, a Jensen zero-counting bound, vertical-strip bounds, and the Nevanlinna bricks toward $\sum_\rho 1/(\frac{1}{4}+\gamma^2) < \infty$.
- `HurwitzRealRootedLimit` — a *from-scratch* Lean formalization of the argument principle (zeros-in-a-contour) and the nowhere-zero

Hurwitz theorem, neither in `mathlib`, giving the headline analytic reduction `det_reg`$\to \Xi \Rightarrow$RH used by the squared-determinant route.

These are the genuinely deep unconditional content of the library and are of independent formalization interest.

6. Methodology and discipline

Axiom hygiene. The headline theorems of all three layers depend on exactly the three standard `mathlib` axioms `[propext,Classical.choice,Quot.sound]`; the tree contains no `sorry` and no added `axiom`. The script `scripts/check_axioms.sh` re-derives this for a curated set of 16 headline names (Appendix 8); it is wrap-tolerant (Lean line-wraps long axiom lists) and fails loudly on any extra axiom, missing name, or `sorry` token. As an independent safeguard, the exported headline declarations and their full dependency closure (54,693 declarations) are re-checked by Nanoda,[1] an independent Rust implementation of the Lean 4 kernel, with the permitted axioms restricted to exactly the three above (`scripts/check_nanoda.sh`). Continuous integration runs the full `lakebuild`, the axiom audit, and the Nanoda re-check on every push.

The falsifier discipline. Every carrier ships with a falsifier: a single non-real regular Ξ-zero refutes its faithful rows. This is not decoration — it forces each interface to be *about* the zeros (an interface satisfiable independently of the zero locations would have no falsifier), and it makes the lattice a genuine reduction rather than a relabeling.

Abstract-interface honesty. Where a module consumes an unconstructed object, its doc-string says so, and a non-supply lemma exhibits a concrete cheap carrier that fails the faithful rows. The library never imports an object it has not built; the open input is always a named, unproven hypothesis.

Production process. The development was produced by AI research agents (Claude "Fable" and GPT) under the author's direction, in a disjoint-file-ownership mutual-audit protocol, with the Lean kernel as final referee — the same process used for Theorem M [5].

7. Honest scope and limitations

We state the limitations plainly, because they determine what the artifact is worth.

(1) **It is not a proof of RH**, and contains none. Every carrier's faithful rows are left unproven; no module instantiates a faithful carrier.

[1] `https://github.com/ammkrn/nanoda_lib`; the export uses `lean4export`, `https://github.com/leanprover/lean4export`.

(2) **The carrier equivalences are, by design, logically shallow.** A carrier is essentially "an object whose existence would prove RH and which a single off-line zero refutes"; proving RH $\iff$ (such an object exists) is not a deep theorem about ζ. Their worth is bookkeeping, mutual consistency, falsifier-faithfulness, and the non-supply lemmas — not depth. We do not claim otherwise.
(3) **The genuinely substantive content is elsewhere:** the no-go certificates (§4), the from-scratch argument-principle/Hurwitz analysis and the Riemann–von Mangoldt bricks (§5), and the reduction spine (§2).
(4) **The single open input,** reached independently by the geometric, spectral, moment, and Morse routes, is one object: a non-spectral geometric carrier (an arithmetic $\operatorname{Spec}\mathbb{Z} \times_{\mathbb{F}_1} \operatorname{Spec}\mathbb{Z}$ / Deninger dynamical cohomology / Connes–Consani arithmetic site) delivering a Hodge-index-positive intersection form, equivalently a faithful zero dictionary. The library names this object and its falsifier; it does not construct it. Constructing it is the open problem.

8. Reproducibility

Build.

```
cd formal && lake build      # green: 128 modules, 8602 jobs
```

Toolchain: `leanprover/lean4:v4.30.0`; `mathlib` pinned at `v4.30.0` (see `lake-manifest.json`).

Axiom + sorry audit.

```
scripts/check_axioms.sh
  -> 16 headline theorems axiom-clean
     ([propext, Classical.choice, Quot.sound]), zero sorries.
```

The audited headline names include the spine (`riemannHypothesis_iff_regular_riemannXi_zeros_real`, `riemannXi_even`, `riemannXiRegularZero_neg`), the lattice (`hasFaithfulHodgeIndexCarrier_iff_riemannHypothesis` and its carrier-to-carrier links, `FredholmSquaredCarrier`, `SpectralRealization`), the no-gos (`not_coprimeMultiplicative_of_DHPattern236`, `ScatteringParityNoGo`, `PrimeLocalNoGo`, `ResolutionWall`), and an analytic brick (`xiE_zero_iff_zeta_zero`).

Sorry sweep.

```
# no real sorry/admit tokens anywhere in the tree:
rg -n '(^|[^A-Za-z_])(sorry|admit)([^A-Za-z_]|$)' \
   formal/JensenLadder
```

9. Related work

Formalized analytic number theory. Substantial parts of the analytic theory of ζ are already formalized. In Isabelle/HOL, Eberl and Paulson formalized a large portion of analytic number theory — including an analytic proof of the Prime Number Theorem and the Riemann/Hurwitz zeta and Dirichlet L-functions [3]. In Lean, the `PrimeNumberTheoremAnd` project (Kontorovich, Tao, and collaborators) formalized the Prime Number Theorem with error term [4], and `mathlib` itself supplies the *statement* `RiemannHypothesis` [2] that this library targets. All of these formalize *theorems about* ζ; none addresses RH's equivalent reformulations or the obstructions to proving it. The present work is complementary — it mechanizes the *logical geometry* around the open statement, not an analytic theorem implying it.

The informal RH landscape. RH has an extensive informal catalogue of equivalent reformulations — two volumes by Broughan [6], Conrey's survey [7], and the official problem description [8] — while the function-field analogue (Weil [9], Deligne) and the noncommutative-geometry program (Connes [10]) supply the "geometric positivity" target that the carriers of §3 abstract. To our knowledge none of this had previously been mechanized; the contribution here is to state these equivalences and obstructions precisely against one kernel, prove them mutually consistent, and pair each carrier with a falsifier and each obstruction with a finite certificate.

10. Conclusion

The Riemann Hypothesis has long had a rich informal "geometry": a web of equivalent reformulations and a folklore of obstructions explaining why the easy routes fail. `jensen-ladder` turns that geometry into a single machine-checked object. The reformulations become a kernel-enforced equivalence lattice; the obstructions become finite, unconditional no-go certificates; the boundary between proved analytic content and abstract reduction target is made explicit at the type level; and the one open input is named, not hidden. None of this proves RH — and the library is scrupulous never to suggest it does. What it offers is a faithful, axiom-clean map of where the problem actually sits, of a kind that, to our knowledge, has not previously been mechanized.

Acknowledgments. The library and its formal verification were developed by AI research agents, Claude ("Fable", Anthropic) and GPT (OpenAI), under the author's direction, with mutual line-audits and the Lean kernel as referee.

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